

Colombo Alessandro

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EDUCATION

From		to	
Sep 2024	-	Apr 2027	**Master of Science in Electronics Engineering** – Politecnico di Milano All exams completed, thesis in progress. Average grade: 23.45/30 **Thesis** *(in progress):* closed-loop verification system for avionics displays. A microcontroller renders the flight symbology (artificial horizon) on an avionics-type display, while optical sensors (photodiodes) check that reference pixels and symbols match the commanded image, raising a warning in case of mismatch.
Sep 2021	-	Sep 2024	**Bachelor of Science in Electronics Engineering** – Politecnico di Milano Thesis: Final Project (Prova Finale) Final grade: 96/110
Sep 2016	-	Jul 2021	**High School Diploma**: **Scientific** – Liceo Scientifico G.B. Grassi (Lecco) Final grade: 76/100

LANGUAGES

Italian: Native speaker
English (UK):

- Written and spoken **B2**: First Certificate in Cambridge (FCE), Toeic Eas Milan (*2021*)
- Written and spoken **C1**: TOEIC, score: 950/990, CEFR Level: C1 (*2024*)

WORK EXPERIENCE

From		to	
Mar 2023	-	Sep 2024	Bartender – Mexicali Restaurant, Lecco
Sep 2023	-	Jun 2024	**Private tutoring** in maths and physics for Liceo Scientifico students
Apr 2022	-	Sep 2022	Pizza delivery driver – Rida Bar&Pizzeria

MAIN COURSES ATTENDED

Bachelor's degree: Mathematical Analysis 1 and 2, Fundamentals of Computer Science, Physics, Electrical Engineering, Electromagnetism and Optics, Fundamentals of Automatic Control, Fundamentals of Electronics, Analog Electronics, Digital Electronic Systems, Fundamentals of Signals, Solid-State Electronics, Electronic Devices.

Master's degree: Electronic systems, Sensor System, Digital Integrated Circuit Design, MEMS and microsensors, Analog Circuit Design, Signal Recovery, Power Electronics, Analog and Digital Holography, Smart Wearables Design and Prototyping, Smart Materials for Wearable, Biochip, Applied AI in Biomedicine, Automation and Control in Autonomous Vehicle.

SKILLS

- **Programming languages:** C, Python, MATLAB, VHDL (basic)
- **Embedded:** STM32 microcontrollers (STM32CubeIDE), Arduino; I²C, SPI and Bluetooth Low Energy protocols; digital microphone acquisition (PDM/MDF)

- **Hardware design:** Altium Designer and KiCad (schematic and PCB layout), PSpice (circuit simulation), Vivado (basic, FPGA implementation)
- **Simulation and control:** MATLAB/Simulink, Vehicle Dynamics Blockset
- **Other:** basic PLC programming (Ladder, Instruction List, SFC – Lovato LRSXW simulator), AutoCAD (basic), Microsoft Office suite
- **Soft skills:** teamwork (university projects), problem solving, time and client management, discipline and consistency, developed also through work experience.

INTERESTS AND PERSONAL INFORMATION

I am very interested in automotive, especially motorsport, and more broadly about systems in which mechanics and electronics work in synergy. I am moving towards the aerospace and defence sectors, also through my thesis on avionics displays, and I am interested in wearable devices, on which I have carried out several projects. I am available for business travel and work experience abroad; in my free time I enjoy sports, photography and travelling.

ACADEMIC PROJECTS

"Computer Generated Hologram Project" – *Analog and Digital Holography* (Politecnico di Milano, individual project): MATLAB generation of an off-axis amplitude hologram from a phase optimized with the Gerchberg–Saxton algorithm. The hologram was printed on transparent film with office printers and reconstructed with a 520 nm laser in the laboratory (*Smart Eyewear Lab – EssilorLuxottica*). By optimizing resolution and off-axis angle, I obtained a sharp reconstruction. By analyzing the diffraction peaks, I then measured the effective resolution of the printers, comparing laser and inkjet printing. Finally, I replicated the experiment at home with a sub-1 mW laser, identifying beam collimation as the limit of the low-cost setup.

"Lumos Probe" – ***Smart Wearables Design and Prototyping*** (Politecnico di Milano, team of 5 students): design and prototyping of a probe for smart eyewear, with an AS7341 spectral sensor, a digital MEMS microphone and an IMU on an STM32U5 microcontroller. I mainly worked on the C firmware (sensor drivers via I²C and MDF, Bluetooth Low Energy transmission) and on the environmental-context classification and user-advice logic in the smartphone app. I also supported the design of the 2- and 4-layer PCB in Altium Designer, handling the sourcing and integration of footprints and schematic symbols.

"Biochip Project" – ***Biochip*** (Politecnico di Milano, team of 5 students): design, fabrication and experimental characterization of a PCB-based liquid impedance sensor, with gold interdigitated coplanar electrodes and a transimpedance front-end (AD8022). The electrode geometry was sized by estimating the solution resistance through conformal mapping. The board was validated through impedance spectroscopy between 1 kHz and 1 MHz with a Moku platform (lock-in), measuring deionized water, tap water and saline solutions.

"Aegis Cerbero" – ***Smart Materials for Wearables*** (Politecnico di Milano, team of 4 students): concept of a smart mouthguard for contact sports that monitors physical stress in real time through salivary alpha-amylase. The system integrates an aptamer-functionalized magnetic hydrogel, an electrospun TPU/PEG membrane and a Hall sensor with a microcontroller, which triggers haptic feedback and transmits data via BLE. The project included material selection, MATLAB modelling, cost analysis (BoM) and limitations.

"Trajectory Tracking Controller" – ***Automation and Control in Autonomous Vehicle*** (Politecnico di Milano, individual project): design in MATLAB/Simulink (Vehicle Dynamics Blockset) of an MPC controller that tracks the position and speed of a vehicle with F1-like parameters over a full lap of the Monza circuit. I developed a global planner for the racing line, which optimizes the length/curvature trade-off and computes the speed profile on the friction ellipse. I then tuned the controller (sampling time, horizons, weights) based on the analysis of simulation data.